The TIPS–Treasury Basis as a Limits-to-Arbitrage Equilibrium

A Structural Model of an Equilibrium Mispricing Floor

Detailed Project Description and Preliminary Results

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Abstract

We ask why the TIPS–Treasury mispricing of Fleckenstein, Longstaff and Lustig [2014]—a default-free synthetic nominal bond (TIPS plus inflation swap) trading cheap to the cash Treasury—persists at a small, non-zero, occasionally negative level after 2013, having reached roughly 150 bps in 2008–09, and what determines that level. The existing literature documents and decomposes the basis; we instead model it. We develop a structural decision-theoretic model in which a risk-averse, capital-constrained arbitrageur enters the trade only if the certainty equivalent of its mark-to-market-risky payoff exceeds the capital charge it consumes. The observed basis is then the level at which the marginal constrained arbitrageur is indifferent: small in calm states (the certainty equivalent falls short of the charge, no one enters, and the spread is pinned at an equilibrium floor), widening in stress states (the forced-unwind tail fattens). The post-2013 compression is thus the model's prediction, not the disappearance of the anomaly. We estimate the structural parameters by simulated method of moments, using the He–Kelly–Manela intermediary capital ratio as the state variable for the capital charge, and deliver policy counterfactuals on balance-sheet constraints. Preliminary analysis of a 2004–2026 panel of 107 individual TIPS is encouraging and internally coherent. The cross-sectional-median basis falls from a pre-2013 mean of 37.6 bps (peak 153.4 bps, November 2008) to a post-2013 mean of 17.7 bps; post-2013 it is a well-defined mean-reverting process (Dickey–Fuller rejects a unit root at the 1% level, half-life 1.9 months, Ornstein–Uhlenbeck long-run level 17.2 bps), it turns negative on 76 days in 2021–2022 (minimum −7.5 bps), and its monthly changes load on the He–Kelly–Manela intermediary capital ratio with the model-predicted sign—insignificant unconditionally but significant (-26.2, $p < 0.05$) in top-quintile-volatility stress months. The mechanism is alive and stress-conditional, exactly as the model requires.

Keywords: Limits to arbitrage, TIPS–Treasury basis, intermediary asset pricing, slow-moving capital, certainty equivalent, structural estimation, simulated method of moments.
JEL Classification: G12, G14, E43, C58.

1. Introduction

A Treasury Inflation-Protected Security (TIPS) whose realised-inflation exposure is removed by an offsetting inflation swap, combined with coupon-matching nominal STRIPS, replicates the cash flows of a default-free *nominal* Treasury. By the law of one price the two must trade at the same price. Fleckenstein, Longstaff and Lustig [2014, henceforth FLL] show that they do not: the synthetic is systematically *cheap*—it offers a higher yield by an amount λ—with the dollar mispricing exceeding \$20 per \$100 notional at its peak and an aggregate above \$56 billion, nearly 8% of TIPS outstanding. FLL attribute the persistence of the anomaly to *slow-moving capital*

[Duffie, 2010], supported by direct evidence that the mispricing narrows as arbitrage capital flows into the market.

The empirical fact that organises this project is the *collapse of λ across regimes.* In our 2004–2026 panel the cross-sectional-median basis averages 37.6 bps before 2013, peaking at 153.4 bps in November 2008; after 2013 it averages 17.7 bps and never exceeds 48 bps, with only muted widening even in March 2020. The naive inference is that the arbitrage has been competed away and there is nothing left to study. We take the opposite position, which is the thesis of the paper:

> *The compression of λ is the signature of binding limits to arbitrage, not evidence against them. Dedicated relative-value capital entered, arbitraged away the easily harvested spread, and stopped precisely where the residual no longer compensates a constrained arbitrageur for the risk of a forced unwind. What survives is an* equilibrium floor*, not a vanished anomaly—and that floor is a quantitative object a structural model can pin down.*

A second fact sharpens the design. After 2013 the basis is not merely small but *bidirectional*: on 76 days in 2021–2022 the cross-sectional median turned negative (TIPS became *rich*, minimum -7.5 bps in May 2021), the opposite sign of the original FLL dislocation and the phenomenon Fleckenstein and Longstaff [2024] study as "Treasury richness." Both signs appear within the same regime, pointing to a symmetric, constraint-driven mechanism rather than a one-sided cheapness anomaly. A satisfactory model should accommodate either sign.

Contributions. The project makes five contributions.

1. **A structural model of the basis *level.*** We move from "the basis is X and correlates with Y" to "the basis is X *because* it is the level at which the marginal constrained arbitrageur is indifferent, given the capital charge." The basis becomes an equilibrium object derived from a certainty-equivalent entry condition, not a measured residual. To our knowledge no existing study writes down the arbitrageur's entry decision for this trade and solves for the basis as the equilibrium indifference level.

2. **The post-2013 compression as an equilibrium floor.** We reframe the under-studied calm regime: the model *generates* a small, non-zero floor as the compensation a constrained arbitrageur requires for forced-unwind risk. A "the arbitrage is gone" non-result becomes a structural statement about the equilibrium level, consistent with FLL's slow-moving-capital evidence (capital entered and stopped at the floor).

3. **A unified, signed treatment of bidirectional dislocation.** The mechanism opens the basis in whichever direction faces the binding constraint, nesting the Fleckenstein (TIPS-cheap) and Treasury-richness (TIPS-rich, 2021) phenomena within one model.

4. **Structural estimation with policy counterfactuals.** Using simulated method of moments to match stress-conditional sensitivities and mean-reversion moments, we estimate the arbitrageur's risk aversion and the capital-charge function, and answer counterfactual questions—how far the basis would compress under a halved capital charge—that reduced-form correlations cannot address.

5. **Methodological bridge from the author's companion studies.** The reduced-form layer extends the cross-strategy framework of the author's study of European fixed-income arbitrages (co-movement, stress amplification, lead–lag, persistence and funding-dependence, common factor) from description to structural explanation, and the intermediary-capital state variable links the present project to the author's parallel work on state-dependent multi-asset allocation, in which a data-driven selection independently recovers intermediary-balance-sheet proxies.

Research questions. The investigation is organised around five questions.

RQ1 (Co-movement).
: Which limits-to-arbitrage state variables—intermediary capital, funding/repo stress, Treasury-market illiquidity, credit—drive the basis, and is their explanatory power *state-dependent*, weak in calm regimes and re-emerging under stress?

RQ2 (Structural identification).
: What values of arbitrageur risk aversion γ and of the capital charge are jointly consistent with the observed equilibrium basis and its stress-conditional dynamics?

RQ3 (Mechanism).
: Is the forced-unwind channel—the negative tail of the payoff distribution—necessary to match the data, or does slow-moving capital without binding unwinding suffice? Equivalently, would a representative arbitrageur have been stopped out in 2008–09 and March 2020 under realistic capital charges?

RQ4 (Counterfactual / policy).
: How would the equilibrium basis respond to a change in the capital charge (e.g. looser or tighter balance-sheet regulation) or in arbitrageur risk aversion?

RQ5 (Sign).
: Can the same mechanism, in a signed specification, account for both the Fleckenstein (TIPS-cheap) and the Treasury-richness (TIPS-rich, 2021) episodes?

Preview of findings. Our preliminary results, on daily data for the level dynamics and monthly data for the factor co-movement, are fourfold. First, the post-2013 basis is a well-behaved mean-reverting spread: the augmented Dickey–Fuller test rejects a unit root at the 1% level (it does not pre-2013), the half-life falls from 2.5 to 1.9 months, and the Ornstein–Uhlenbeck long-run level is 17.2 bps—a candidate empirical counterpart of the equilibrium floor. Second, the basis is genuinely bidirectional after 2013 (76 negative days, minimum -7.5 bps), so a signed specification is required. Third, the He–Kelly–Manela intermediary capital ratio—the empirical proxy for the model's capital charge—is insignificant unconditionally post-2013 but significant and correctly signed in stress months (-26.2, $p < 0.05$), and prime-broker CDS, Amihud illiquidity and CDX IG behave likewise: the mechanism is alive and stress-conditional. Fourth, the identification challenge is real and we are explicit about it: only two episodes (the 2008–09 crisis and the June 2013 taper tantrum) clear a regime-relative three-sigma threshold, March 2020 reaches only about two sigma at the median, and the single largest structural break in the level is dated 2010, not

2013—facts that motivate the planned daily factor reconstruction and the use of the bidirectional 2021 episode for identification.

Roadmap. Section 2 states the economic motivation and the falsifiable hypotheses. Section 3 develops the structural model. Section 4 describes the data; Section 5 the methodology. Section 6 reports preliminary results; Section 7 discusses them; Section 8 lays out the remaining work; Section 9 concludes.

2. Economic Motivation and Hypotheses

The case for a structural treatment of the basis rests on three connected pillars.

2.1 The no-arbitrage identity and the puzzle

The TIPS–Treasury trade is the cleanest near-arbitrage in fixed income: a long position in the inflation-swapped TIPS and a short in the maturity-matched nominal Treasury replicates a zero net cash flow at maturity, so any non-zero spread λ is, in a frictionless world, a violation of the law of one price. FLL establish that λ is large, persistent and positive, and that it co-moves with supply factors (Treasury issuance, collateral availability) and with other fixed-income arbitrages. The puzzle is not that λ exists in a crisis—frictions obviously bind then—but that it survives, small and positive, for a decade afterwards. That survival is what a structural model must explain.

2.2 Limits to arbitrage and intermediary capital

Arbitrage requires capital; that capital is supplied by specialised, leveraged intermediaries; and their constraints bind precisely when mispricings are largest [Shleifer and Vishny, 1997; Gromb and Vayanos, 2002, 2010; Brunnermeier and Pedersen, 2009; Duffie, 2010; Garleanu and Pedersen, 2011; Kondor and Vayanos, 2019]. Intermediary asset pricing makes the intermediary's marginal value of wealth the relevant pricing kernel and supplies an empirical proxy—the equity capital ratio of the New York Fed's primary dealers [He and Krishnamurthy, 2013; He, Kelly and Manela, 2017]. When intermediary equity falls, or funding costs and volatility rise, intermediaries deleverage, the effective risk aversion of the marginal arbitrageur rises, and the compensation demanded to hold a mark-to-market-risky convergence trade increases. The empirical implication is sharp: the basis should co-move with credit spreads, dealer-funding stress and intermediary capital, and the co-movement should intensify when constraints bind. Our capital charge is the operational, trade-level counterpart of this constraint, and we use the He–Kelly–Manela ratio as its state variable.

2.3 The mark-to-market channel, forced unwinding, and the equilibrium floor

The distinctive feature of the trade—and the reason a simple "free profit" framing fails—is that, although the terminal payoff is locked in, the position is marked to market and can run a deep paper loss before convergence (Section 3). A leveraged arbitrageur against a capital charge is forced to unwind at a loss if the drawdown breaches a threshold. The realised payoff is therefore a mixture of a convergence profit and a forced-liquidation loss, and a risk-averse investor prices its certainty equivalent against the capital the trade consumes. The basis settles at the level that makes the marginal constrained arbitrageur indifferent: small in calm states (a floor), wide in stress states. This is the mechanism we formalise and estimate.

2.4 Falsifiable hypotheses

The discussion yields three falsifiable hypotheses; in each case the null is that the data are indistinguishable from the stated alternative's negation.

Hypothesis 1 (Equilibrium floor)**.** *The post-2013 basis is a stationary, mean-reverting process with a strictly positive long-run level that is small relative to the pre-2013 regime. The null—a unit root, or a long-run level statistically indistinguishable from zero or from the pre-2013 level—would reject the equilibrium-floor interpretation in favour of either a non-stationary residual or full convergence to no-arbitrage.*

Hypothesis 2 (Stress-conditional capital channel)**.** *The sensitivity of the basis to intermediary-capital, funding and illiquidity state variables is state-dependent: weak or absent in calm regimes and significant, with the sign predicted by the constraint mechanism, in stress regimes. The null—no co-movement with these variables in any regime, or co-movement indistinguishable across regimes—would reject the limits-to-arbitrage mechanism in favour of a frictionless or purely macro-driven account.*

Hypothesis 3 (Binding forced-unwind)**.** *Under realistic capital charges, a representative convergence trade would have been forced to unwind at a loss in historical stress episodes, and a structural model that includes the forced-unwind tail matches the level and dynamics of the basis better than one with slow-moving capital alone. The null—that the trade is never stopped out under plausible charges, so the forced-unwind tail is empirically irrelevant—would reduce the mechanism to pure slow-moving capital.*

3. The Structural Model

We follow the construction of Rebonato. Upper-case symbols denote nominal quantities (P_0^T, Y_0^T for the price and yield of a T-maturity nominal zero), lower-case the corresponding real quantities (p_0^T, y_0^T); f_0^T is the forward inflation compensation and $\pi(t)$ instantaneous inflation.

3.1 The synthetic nominal bond and the law of one price

Consider three instruments: a zero-coupon TIPS, an inflation swap (IS), and a nominal bond. A unit of TIPS pays at T

$$\exp\left[\int_0^T \pi(s)\,ds\right], \tag{1}$$

and a unit of the IS pays

$$\exp\left[f_0^T(T-t_0)\right] - \exp\left[\int_0^T \pi(s)\,ds\right]. \tag{2}$$

The portfolio Π_0^T of one TIPS plus one IS therefore pays a *certain* amount, identical in every state of the world,

$$\Pi_0^T = \exp\left[f_0^T(T-t_0)\right]. \tag{3}$$

Since a T-maturity nominal zero pays \$1 with certainty, $\exp[-f_0^T T]$ units of the TIPS+IS combination also pay \$1 with certainty, so under no constraints the law of one price requires

$$\exp\big[-f_0^T T\big]\, p_0^T = P_0^T \quad \Longrightarrow \quad p_0^T = P_0^T \exp\big[-f_0^T T\big]. \tag{4}$$

Writing $P_0^T \equiv \exp[-Y_0^T T] = \exp[-(y_0^T + f_0^T)T]$ recovers the consistency condition

$$p_0^T = \exp\big[-y_0^T T\big], \tag{5}$$

so the synthetic portfolio Π_0^T is a security whose payoff matches the nominal zero, and under no arbitrage $Y(\Pi_0^T) = Y_0^T$.

3.2 The mispricing

Suppose instead the synthetic trades away from parity,

$$\Pi_0^T \neq P_0^T, \tag{6}$$

reflected in a yield wedge

$$Y(\Pi_0^T) = Y_0^T + \lambda_0^T, \qquad \lambda_0^T \neq 0. \tag{7}$$

The scalar λ_0^T is the basis: the FLL puzzle is $\lambda_0^T > 0$ (Treasuries rich / synthetic cheap); the Treasury-richness episodes correspond to $\lambda_0^T < 0$. Were Π_0^T a derivative, the long-synthetic / short-nominal position could be "upfronted" for an immediate profit

$$PV(0) = \exp\big[-Y_0^T T\big]\Big(1 - \exp\big[-\lambda_0^T T\big]\Big). \tag{8}$$

3.3 Mark-to-market dynamics and forced unwinding

In practice every leg is a traded security marked to market, so the strategy (long synthetic, short nominal, with the net cash invested at Y_0^T) has *zero* value at inception and accrues value through time. With $\tau \equiv T - t$, the time-t value of the whole strategy is

$$S(t) \;=\; \underbrace{\exp\big[-Y_0^T \tau\big]\Big(1 - \exp\big[-\lambda_0^T \tau\big]\Big)}_{\text{deterministic (pull-to-convergence)}} \;-\; \underbrace{\exp\big[-Y_t^T \tau\big]\Big(1 - \exp\big[-\lambda_t^T \tau\big]\Big)}_{\text{stochastic (mark-to-market)}}. \tag{9}$$

The first term is deterministic; the second is driven by the realised paths of the nominal yield Y_t^T and the basis λ_t^T. The economic content of (9) is the crux of the project: **the strategy is worth more when nominal yields rise and the spread narrows, and less when yields fall and the spread widens**. An instantaneous jump in the basis at inception, from λ_0^T to $\hat{\lambda}_0^T$, moves the strategy by

$$\hat{S}(0) = \exp\big[-Y_0^T T\big]\Big(\exp\big[-\hat{\lambda}_0^T \tau\big] - \exp\big[-\lambda_0^T \tau\big]\Big), \tag{10}$$

negative precisely when the basis widens. Held to maturity the terminal profit is locked in at $P_0^T\big(1 - \exp[-\lambda_0^T T]\big)$; *before* maturity the position can run an arbitrarily deep paper loss. A leveraged arbitrageur against a capital constraint cannot absorb an unbounded drawdown:

when the cumulated mark-to-market loss breaches a threshold, the position is unwound by force, crystallising the loss. The realised payoff is therefore a *mixture distribution*:

1. a Dirac mass at the positive convergence profit $P_0^T\big(1 - \exp[-\lambda_0^T T]\big)$, with probability equal to the fraction of paths that reach T without a forced unwind;

2. a continuous *negative tail*, each value equal to the discounted strategy value at the forced-unwinding time, over the paths that are stopped out.

3.4 The entry decision and the equilibrium

The mixture distribution has positive expectation, but a risk-averse investor does not price the mean. She prices the certainty equivalent of her terminal wealth and enters only if it exceeds her initial wealth, which following Rebonato we identify with the *capital charge* the strategy ties up:

$$\text{enter} \quad \Longleftrightarrow \quad \mathrm{CE}_\gamma\big(W_0 + \widetilde{S}_T\big) > W_0, \qquad W_0 = m \cdot \mathrm{VaR}, \tag{11}$$

where $\widetilde{S}_T$ is the mixture-distributed realised payoff, γ the coefficient of (constant relative) risk aversion, and m a regulatory/risk multiplier. The conceptually correct charge is the *marginal* VaR contribution of the trade; the full VaR is a tractable approximation. To close the model we specify a mean-reverting process for the basis,

$$d\lambda_t = \kappa\,(\theta - \lambda_t)\,dt + \sigma\,dW_t, \tag{12}$$

with the capital charge W_0 a function of an intermediary-distress state variable (empirically, the He–Kelly–Manela capital ratio). The observed basis is the level at which the marginal constrained arbitrageur is indifferent in (11), which produces the regime behaviour directly:

- **Calm states.** The charge is small, but so are λ and the volatility of $S(t)$; the certainty equivalent falls short of the charge; no constrained arbitrageur enters; λ is pinned at a small, non-zero *equilibrium floor*.

- **Stress states.** Nominal yields fall, volatility and funding costs rise, intermediary capital contracts; the charge spikes and the forced-unwind tail fattens; the indifference level—and hence λ—widens.

A small, persistent, occasionally exploding and occasionally sign-flipping basis is exactly what the model predicts. Crucially, in the structural model the reduced-form regression loadings of Section 6 are *not* free parameters: they are implied functions of $(\gamma, m, \kappa, \theta, \sigma)$, so the model is asked to reproduce them.

3.5 Open modelling choices

Three decisions are left open and will be settled before estimation.

- **Initial wealth:** full VaR versus marginal VaR as W_0 in (11); the marginal notion is economically correct but harder to operationalise, with a desk-calibrated charge as a pragmatic third option.

- **Sign of λ:** a signed specification admitting the 2021 negative episode, versus absolute deviation; the bidirectional evidence argues for the signed version (RQ5).

- **State dependence of the charge:** whether W_0 is a smooth function of the capital ratio or a regime indicator.

4. Data and Factor Universe

4.1 The basis

The basis is constructed from a panel of 107 individual TIPS identified by CUSIP, daily, from 2 August 2004 to 18 May 2026 (5,446 trading days), each cell the security-level basis in bps relative to the maturity-matched nominal Treasury. The number of live securities grows from about 14 in 2004 to about 52 in 2024–26. The working aggregate is the daily cross-sectional median of active CUSIPs; the aggregation choice—median, maturity-bucketed, or a single representative on-the-run 10-year "representative trade"—is itself a modelling decision, and the representative-trade aggregate is the natural input to the structural simulation of Section 5.3.

4.2 Rates, inflation and market stress

From Bloomberg, daily: nominal yields (2-, 5-, 10-, 30-year), the 1-month bill, the 5y5y USD inflation-swap forward, the 5-year breakeven, VIX, MOVE, the S&P 500 total-return index, and Treasury and TIPS total-return indices for the mark-to-market simulation.

4.3 Limits-to-arbitrage factor universe

Reused from the author's companion fixed-income study, organised by economic channel:

1. **Intermediary capital:** the He–Kelly–Manela intermediary capital ratio, and prime-broker CDS (1- and 5-year).
2. **Treasury-market illiquidity:** the Hu–Pan–Wang noise measure, the Amihud illiquidity measure, and primary-dealer fails-to-deliver.
3. **Funding:** the GC repo–T-bill spread and the LIBOR–OIS spread (SOFR-based after June 2023).
4. **Credit:** the CDX investment-grade index and its option-implied volatility, and the Gilchrist–Zakrajšek excess bond premium.
5. **Rates and risk-off:** the change in the 10-year yield, MOVE, VIX, and the S&P total return.
6. **Inflation:** the 5y5y inflation swap and the 5-year breakeven.

4.4 Frequency and no-look-ahead

The level dynamics (stationarity, mean reversion, episodes) are studied at *daily* frequency. The factor co-movement is currently at monthly frequency because the companion-study factor files are end-of-month; monthly aggregation damps short-lived intra-month spikes such as 9–23 March 2020—the same averaging that flattens the dislocation in Ahn and Ahn [2023]. Because the forced-unwind mechanism lives in those spikes, the first task of the full study is a daily reconstruction of the factor set. All slow-released series (the capital ratio, the excess bond premium, fails) are aligned to release dates; market-observable series are taken at the close.

5. Methodology

The empirical design proceeds in three blocks—the first two reduced-form, establishing the moments the structural model must match, the third structural—followed by inference.

5.1 Constructing and validating the basis

We reconstruct λ_t at daily frequency from the FLL replication (TIPS plus a maturity-matched zero-coupon inflation swap plus coupon-matching STRIPS), rather than from a fitted-curve residual, so that λ is an executable, self-financing spread. We test stationarity with the augmented Dickey–Fuller (AIC lag) and KPSS (automatic lag) tests, with Phillips and Perron [1988] as a robustness check; estimate an AR(1)/Ornstein–Uhlenbeck specification yielding (κ, θ, σ) and the half-life $\ln 2/\kappa$; and bound the no-arbitrage band by comparing λ to a round-trip transaction-and-funding cost (bid–ask on the TIPS, the inflation swap and STRIPS, plus repo specialness and haircut financing). We date regime shifts with the Bai and Perron [1998, 2003] multiple-break procedure and sup-F/UDmax statistics on the level and the volatility of λ (15% trimming).

5.2 Co-movement and the mechanism (RQ1, RQ5)

We regress weekly and monthly changes in λ on each limits-to-arbitrage state variable, one at a time, separately for the pre-2013, post-2013 and stress (top-quintile-VIX) samples, with Newey–West HAC inference. Quantile regressions [Koenker and Bassett, 1978] at $\tau = 0.5$ and 0.95 capture the asymmetric tail response, and a small vector autoregression in $(\lambda, Y, \mathrm{VIX}, \mathrm{funding})$ with generalised impulse responses [Pesaran and Shin, 1998] characterises lead–lag transmission and the slow post-shock reversion that identifies slow-moving capital. This block deliberately mirrors the cross-strategy propositions of the companion study—co-movement, stress amplification, lead–lag, persistence and funding-dependence, common factor—recast as inputs to the structural step.

5.3 Structural estimation (RQ2–RQ4)

We simulate (λ_t, Y_t) under (12); for a representative entry, replay the mark-to-market value (9) and impose a forced unwind when the cumulated loss breaches the capital charge $W_0 = m \cdot \mathrm{VaR}$, with W_0 keyed to the intermediary capital ratio; and compute the mixture-distributed payoff and the certainty equivalent (11). We estimate $\eta = (\gamma, m, \kappa, \theta, \sigma)$ by simulated method of moments [McFadden, 1989; Duffie and Singleton, 1993], with target moment vector drawn from blocks 5.1–5.2: the stress-conditional sensitivity of λ to intermediary capital, the mean-reversion half-life, the unconditional mean and volatility of λ, and the frequency and depth of widening episodes. The weighting matrix is the inverse of the block-bootstrap [Politis and Romano, 1994] covariance of the empirical moments, and we report the J-test of over-identifying restrictions. Validation proceeds by checking that the estimated model reproduces the out-of-sample calm regime (a small floor with no entry); the capital-charge and risk-aversion counterfactuals (RQ4) then vary m and γ and trace the implied equilibrium basis.

5.4 Inference

Throughout we use Newey and West [1987] HAC standard errors; risk-adjusted relationships are additionally estimated in a conditional specification à la Ferson and Schadt [1996] that lets the loadings vary with the funding-stress state, and the structural fit is assessed by the SMM J-statistic and by out-of-sample moment reproduction.

6. Preliminary Results

We report preliminary evidence on the cross-sectional-median basis: the level dynamics on daily data (2004–2026, 5,446 observations), the factor co-movement on monthly data. Throughout, the regime split is set at 1 January 2013 to isolate the stable low-basis era; the data-driven dating is discussed in Section 6.3.

6.1 Descriptive statistics by regime

Table 1 reports the distribution of the basis by regime. The compression is stark: the mean falls from 37.6 bps before 2013 to 17.7 bps after, the standard deviation from 23.2 to 8.2 bps, and the maximum from 153.4 to 48.0 bps. Two features matter for the model. First, the pre-2013 distribution is strongly right-skewed (3.16) and fat-tailed (excess kurtosis 14.7), whereas the post-2013 *level* distribution is near-symmetric (skewness −0.21) and thin-tailed—the large right tail is a pre-2013 phenomenon. Second, the basis turns negative on 2.3% of post-2013 days (it is strictly positive on every pre-2013 day), the bidirectional evidence that motivates the signed specification.

Statistic (bps)	Full	Pre-2013	Post-2013
Observations	5,446	2,105	3,341
Mean	25.4	37.6	17.7
Median	22.5	31.9	18.2
Std. deviation	18.5	23.2	8.2
Minimum	−7.5	4.5	−7.5
Maximum	153.4	153.4	48.0
Skewness	3.16	2.72	−0.21
Excess kurtosis	14.7	7.9	0.2
% days > 0	98.6	100.0	97.7
AR(1) (daily)	0.990	0.987	0.982

Table 1: Distribution of the cross-sectional-median TIPS–Treasury basis by regime, daily, 2004–2026.

6.2 Mean reversion (Hypothesis 1)

Table 2 supports the equilibrium-floor interpretation. Post-2013 the augmented Dickey–Fuller test rejects a unit root at the 1% level ($p = 0.008$), whereas pre-2013 it does not ($p = 0.135$)—the pre-2013 series is dominated by the slow 2008–09 mean-reversion. The half-life falls from 2.5 to 1.9 months, and the Ornstein–Uhlenbeck long-run level is 17.2 bps post-2013 against 37.4 bps pre-2013. The post-2013 basis thus behaves as a well-defined, faster-reverting spread around a strictly positive level near 17 bps—a natural empirical counterpart of the model's floor. (The KPSS test rejects level-stationarity in every sample, the familiar tension that arises for highly persistent series; we will resolve it with the Phillips–Perron and multiple-break analysis.)

	Full	Pre-2013	Post-2013
ADF p-value (H_0: unit root)	0.016	0.135	0.008
AR(1) coefficient	0.990	0.987	0.983
Half-life (days)	68.4	51.7	39.5
Half-life (months)	3.3	2.5	1.9
OU mean-reversion speed κ	2.55	3.38	4.42
OU long-run level θ (bps)	24.9	37.4	17.2
OU volatility σ	42.0	60.0	24.7

Table 2: Stationarity and mean reversion of the basis, daily. κ, θ, σ are annualised Ornstein–Uhlenbeck parameters; half-life $= \ln 2/\kappa$.

6.3 Structural break and episode inventory

A single-break sup-F scan dates the largest level break at September 2010 (sup-F = 2,533) and the largest volatility break at November 2009—the post-GFC normalisation, *not* 2013. We therefore treat the 2013 split as a pragmatic marker of the stable low-basis regime rather than as the dominant break, and the planned multiple-break Bai–Perron analysis is expected to date both the 2010 normalisation and a later stabilisation. A regime-relative three-sigma threshold (101 bps pre-2013, 43 bps post-2013) identifies only two episodes: the 2008–09 crisis (28 October 2008 to 14 April 2009, 169 days, peak 153.4 bps) and the June 2013 taper tantrum (5 days, peak 48.1 bps). March 2020 reaches 34.6 bps at the median—about two sigma, below the post-2013 threshold—and the 2022–23 inflation episode peaks at 28.0 bps. The identification of the structural model thus leans heavily on the single 2008–09 episode at the median; this is an honest limitation (relevant to RQ3) that motivates both the daily reconstruction, which will sharpen the crisis windows, and the explicit use of the bidirectional 2021 episode (Table 1) as additional identifying variation.

6.4 Co-movement with limits-to-arbitrage state variables (Hypothesis 2)

Table 3 regresses the monthly change in the basis on each state variable, separately by regime and in stress months (top-quintile VIX); $n = 100$ (pre), 161 (post), 32 (post-stress). The pattern is the one the mechanism predicts. The He–Kelly–Manela intermediary capital ratio—the empirical proxy for the capital charge—is insignificant unconditionally post-2013 but significant and correctly signed in stress months (-26.2, $p < 0.05$); prime-broker CDS, Amihud illiquidity and CDX IG are significant post-2013 and strengthen in stress; and the pre-2013 loadings are uniformly larger, the same mechanism at higher amplitude. The inflation channel (breakeven, 5y5y swap) is significant post-2013 with a *negative* sign—when inflation expectations rise, TIPS are bought for hedging, become rich, and the basis compresses or inverts, the 2021 episode in coefficient form (RQ5).

6.5 Honest caveats

Three limitations are worth stating plainly. *Rates insensitivity:* the change in the 10-year yield is insignificant even in stress, plausibly because the two legs are near DV01-matched by construction, muting the direct rate exposure in (9). *Few extreme episodes:* at the median, only 2008–09 and the June 2013 taper tantrum clear three sigma, so the structural identification relies

Factor (channel)	Pre-2013	Post-2013	Post | stress	Exp.
Intermediary capital				
HKM intermediary capital ratio	-74.2^{***}	-8.75	-26.2^{**}	$-$
Prime-broker CDS (5y)	$+0.02$	$+0.12^{**}$	$+0.24^{***}$	$+$
Prime-broker CDS (1y)	$+0.00$	$+0.17^{**}$	$+0.32^{**}$	$+$
Treasury-market illiquidity				
Amihud illiquidity	$+24.7^{**}$	$+5.09^{*}$	$+11.9^{***}$	$+$
Hu–Pan–Wang noise	$+4.59^{***}$	$+1.24$	$+2.97$	$+$
Primary-dealer fails (% debt)	$+11.8$	$+28.0$	$+54.3$	$+$
Credit stress				
CDX IG	$+0.29^{**}$	$+0.14^{**}$	$+0.21^{***}$	$+$
CDX-IG option-implied vol.	$+0.47^{**}$	$+0.13^{**}$	$+0.11$	$+$
Funding				
LIBOR–OIS	$+16.0$	$+3.00$	$+6.24^{*}$	$+$
GC repo–T-bill	$+0.39$	$+2.47$	$+5.54$	$+$
Rates / risk-off				
Δ 10-year yield	-8.73	-0.10	-1.04	$-$
Δ VIX	$+0.72$	$+0.15^{*}$	$+0.17$	$+$
S&P total return	-1.10^{**}	-0.29^{**}	-0.51^{***}	$-$
Inflation				
Δ 5y5y inflation swap	$+14.4$	-13.0^{**}	-18.5^{*}	?
Δ 5-year breakeven	-17.4^{**}	-5.24	-9.96^{**}	?

Table 3: Univariate regressions of the monthly change in the basis on each state variable, by regime and in stress months. HC1 inference; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Slopes in bps of basis per unit of factor.

heavily on the 2008–09 episode and on the bidirectional 2021 variation; the daily reconstruction is needed to recover the March-2020 spike that monthly data smooth away. *Imperfect placebo:* a European credit index (iTraxx Main) remains significant, reflecting the global co-movement of credit, although European rates and Euribor–OIS are clean. The structural step must therefore be disciplined by out-of-sample moment reproduction rather than by in-sample fit alone.

7. Discussion

The preliminary evidence is internally coherent and, taken together, supports the equilibrium-floor reading rather than the "arbitrage is gone" reading. The post-2013 basis is stationary and faster-reverting around a small positive level (17 bps), exactly the floor the model generates; the level distribution is thin-tailed and near-symmetric, yet daily changes remain highly leptokurtic (excess kurtosis 23.4), so the jump risk the forced-unwind mechanism requires is present even in the calm regime. The capital, illiquidity and credit channels co-move with the basis with the predicted sign, and—decisively for the structural project—the intermediary capital ratio becomes significant precisely in stress states, when the model says the charge spikes and the forced-unwind tail fattens. The bidirectional 2021 episode shows that the dislocation can open in either direction, consistent with a symmetric constraint mechanism and with Fleckenstein and Longstaff's [2024] Treasury-richness phenomenon.

The findings also connect the author's research programme. In the companion study of European fixed-income arbitrages, the common factor driving the BTP Italia, CDS–bond and iTraxx

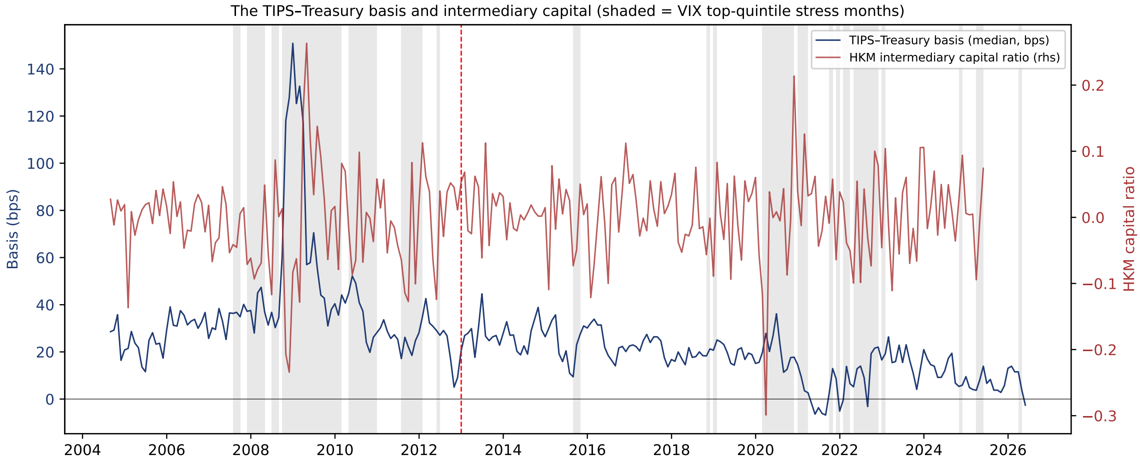

Figure 1: The cross-sectional-median TIPS–Treasury basis (left axis) and the He–Kelly–Manela intermediary capital ratio (right axis), monthly, 2004–2026. Shaded bands are top-quintile-VIX stress months; the dashed line marks the 2013 split. The basis widens as intermediary capital contracts, and the co-movement concentrates in the shaded stress windows.

mispricings loads on intermediary-balance-sheet variables, not on macroeconomic fundamentals; in the parallel study of state-dependent multi-asset allocation, a data-driven selection independently recovers intermediary-capital proxies as the relevant conditioning information. The same latent state—the health of intermediary balance sheets—appears to drive fixed-income mispricings, conditional expected returns, and now the TIPS–Treasury basis. A structural intermediary-asset-pricing model of the kind proposed here is the natural device to unify the three.

8. Roadmap and Next Steps

Four extensions bring the project to the standard of a top-tier submission.

1. **Daily factor reconstruction** of the limits-to-arbitrage set, preserving intra-month spikes (2008–09, March 2020), and collection of the one remaining series (the USD 10-year swap spread, with the 5- and 2-year tenors for robustness); re-estimate Table 3 at daily/weekly frequency and complete the Bai–Perron multiple-break dating. *(Months 1–3.)*

2. **Reduced-form completion:** the no-arbitrage band, the quantile-regression tail analysis, and the VAR/GIRF lead–lag and persistence evidence, delivering the full moment vector for estimation. *(Months 3–5.)*

3. **Structural calibration and estimation:** OU dynamics for λ, mark-to-market replay (9) with a forced-unwind rule keyed to the intermediary capital ratio, certainty-equivalent evaluation (11), and SMM estimation of $(\gamma, m, \kappa, \theta, \sigma)$ with the J-test. The first economic question: at the intermediary-distress levels of 2008–09 and March 2020, does the certainty equivalent exceed the capital charge? *(Months 5–9.)*

4. **Counterfactuals and robustness:** the capital-charge and risk-aversion counterfactuals (RQ4), and robustness to the aggregation choice, the charge definition, the sign specification, and the placebo battery. *(Months 9–12.)*

9. Conclusion

This project asks why the TIPS–Treasury mispricing persists, small and occasionally negative, a decade after the crisis in which it was first measured, and what determines its level. We propose to answer the question structurally: the basis is the level at which a risk-averse, capital-constrained arbitrageur is indifferent between entering the convergence trade and abstaining, given the mark-to-market risk of a forced unwind and the capital charge the trade consumes. The compression after 2013 is then the model's central prediction—an equilibrium floor, not a vanished anomaly. Preliminary analysis of a 2004–2026 panel of individual TIPS is encouraging and internally coherent: the post-2013 basis is a stationary, faster-reverting spread around a 17 bps floor; it is genuinely bidirectional; and it co-moves with intermediary capital, illiquidity and credit with a sensitivity that re-emerges sharply in stress, exactly as the mechanism requires. The remaining work—a daily factor reconstruction, the full reduced-form moment battery, structural estimation by simulated method of moments, and policy counterfactuals—is set out in Section 8.

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